If the polynomial equations are set up with some care, and variables eliminated until you have the promised polynomial equation in a single variable, then the best you can do is to be faced with a cubic. You might think that it would be possible to do better, and get a linear or quadratic equation. In fact that cannot be done, for reasons that will be explained in the Afterword.

There is a theory of cubic equations that enables you to write down their roots. However, you are not required to know this arcane procedure for a Mathematical Olympiad. Rather you are supposed to spot a root t, divide through by an appropriate linear factor, and then look at the resulting quadratic.

Solution

Method We will incorporate a great deal of symmetry into the way we set up the equations. This will pay dividends in making the algebra much easier. Let the product of the radii of the cylinders be r^2, so the radii of the cylinders will be rt_1 and rt_2 where $t_1t_2 = 1$. This description of the radii is symmetric. You cannot tell which cylinder has the larger radius by studying this arrangement. Notice that r is the geometric mean of the radii of the cylinders. We do all this to avoid favouritism. The more symmetry we build into the equations, the easier they should be to handle.

Let the corresponding cylinder heights be h_1 and h_2, where $h_1 + h_2 = 1$, $h_1 = krt_1$ and $h_2 = krt_2$. The constant k is there because the cylinders are similar; let $k = \frac{h_1}{rt_1} = \frac{h_2}{rt_2}$. Introduce $x = t_1 + t_2$, also to avoid favouritism. After all, we know $t_1t_2 = 1$, so if we can work out x, then t_1 and t_2 can be recovered as the roots of $y^2 - xy + 1$, a quadratic in y.

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The height condition is that

$$krx = 1. \tag{1}$$

The surface area condition is that

$$2\pi rt_1 \cdot krt_2 + 2\pi rt_1 \cdot krt_2 + 2\pi r^2(t_1^2 + t_2^2) = 8\pi$$

which simplifies to

$$kr^2(t_1^2 + t_2^2) + r^2(t_1^2 + t_2^2) = 4.$$

This says

$$(1+k)r^2(t_1^2 + t_2^2) = 4$$

which reduces to

$$(1+k)r^2(x^2 - 2) = 4. \tag{2}$$

The volume condition is that $\pi r^2 \cdot (krt_1^3 + krt_2^3) = 2\pi$, or equivalently $r^3(x^3 - 3x) = 2$ which, by the height condition, is

$$r^2(x^2 - 3) = 2. \tag{3}$$

Trading off equations (2) and (3) we obtain

$$(1+k)(2+r^2) = 4. \tag{4}$$

Squaring equation (1) and using equation (3) we find that

$$k^2(3r^2 + 2) = 1. \tag{5}$$

Next notice that equation (4) rearranges to

$$r^2 = \frac{2(1-k)}{1+k}. \tag{6}$$

We can eliminate r^2 from equations (5) and (6) to obtain

$$k^2\left(\frac{6(1-k)}{1+k} + 2\right) = 1$$

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which simplifies to

$$4k^3 - 8k^2 + k + 1 = 0.$$

Spotting that $1/2$ is a root of this cubic in k, we can factor it as

$$(2k-1)(2k^2 - 3k + 1) = 0 \tag{7}$$

which gives rise to three roots: $k = \frac{1}{2}$ or $\frac{3\pm\sqrt{17}}{4}$.

We can discount the negative candidate, and $(1+k)(2+r^2) = 4$ so $2 + r^2 = \frac{4}{1+k}$ so $k \leqslant 1$ and we can also discount the other root of $2k^2 - 3k + 1$.

Thus $k = \frac{1}{2}$ is the only viable candidate. We have $(1 + k)(2 + r^2) = 4$ so $r = \sqrt{\frac{2}{3}}$. Now $krx = 1$ so $x = \sqrt{6} = t_1 + t_2$. Therefore t_1 and t_2 are the roots of $y^2 - \sqrt{6}y + 1$ and these are $\frac{\sqrt{6}\pm\sqrt{2}}{2}$.

The candidates for the radii are rt_1 and rt_2 i.e.

$$\sqrt{\frac{2}{3}} \cdot \left(\frac{\sqrt{6} \pm \sqrt{2}}{2} \right)$$

and tidying up these are

$$\frac{3 \pm \sqrt{3}}{3}.$$

ow $k = \frac{1}{2}$, so the corresponding heights are

$$\frac{3 \pm \sqrt{3}}{6}.$$

g these numbers back in to the original equations and they work!

rword There are complex solutions, with the radii of the cylin- being non-real complex numbers. There is no way to dispose of solutions at the outset, since when doing polynomial algebra, not tell the polynomials that you are uninterested in non-real solutions. The only thing